AEM 7500: Resource Economics

Problem Set 1, due October 29, 2026

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I. Static Pollution Game

Two neighboring countries i: Red (R), White (W). In each period decide how much to emit: $a_i \in \{0\,(low), 1\,(medium), 2\,(high)\}$. Publicly observable costs of abatement. Pollution dissipates after each period. Period utility function:

$$U_i(a, s, \varepsilon_i; \theta) = \begin{cases} -\beta_0 a_i - \beta_{1i} a_i a_{-i} - \beta_2(2 - a_i)s_i + \varepsilon_i(a_i) & \text{if } a_i > 0 \\ -\beta_{3i} a_{-i} + \varepsilon_i(ai) & \text{if } a_i = 0 \end{cases}, \; \theta = (\beta_0, \beta_{1R}, \beta_{1W}, \beta_2, \beta_{3R}, \beta_{3W}) \text{ to be estimated, } \varepsilon_i \sim$$

EV. Then $U_i(a_i = 1) = -\beta_0 - \beta_{1i} a_{-1} - \beta_2 s_i$, $U_i(a_i = 2) = -2\beta_0 - 2\beta_{1i} a_{-i}$.

1. Interpret utility function

Interpretation: β_0 is the marginal change in utility for both countries due to own pollution emitted. β_{1R} is the strategic interaction parameter for Red; if < 0, Red seed pollution for each country as strategic complements (race to the bottom), if > 0, as strategic substitutes. Analogous is β_{1W} for strategic interaction from White's perspective. β_2 is the marginal change in utility of a dollar of abatement cost, the same for both countries. β_{3R} is the marginal change in utility for Red country if it does not emit due to the other country's emissions in that period, β_{3W} is analogous for White.

2. Static game

It is static because nothing carries over from one period to the next. Pollution dissipates at the end of the period, so emitting today does not raise the stock of pollution tomorrow, and the abatement cost s_i is drawn period by period rather than being affected by past decisions. There is no state variable linking periods and therefore no continuation value: each country solves the same one-period problem in every period. It is a game because each country's payoff depends on the other country's action, not only its own. The neighbor's emission a_{-i} enters U_i through $-\beta_{1i} a_i a_{-i}$ when $a_i > 0$ and through $-\beta_{3i} a_{-i}$ when $a_i = 0$, so pollution traveling across the border is an externality that Red imposes on White and White imposes on Red. Neither country can choose its own emission without forming a belief about what the other will do.

3. Estimate choice probabilities nonparametrically

In `PS2_Q3.m` I estimate the conditional choice probabilities nonparametrically as empirical frequencies. In lines 17–18, I read the data and split it into the 374×2 action matrix a, containing pollution emitted by Red and White, and the 374×2 state matrix s, containing the abatement costs in Red and White. In line 22, I call `ccp_nonparam.m`, which does the estimation. In `ccp_nonparam.m`, I condition on the full state pair (s_R, s_W) rather than on own cost alone, since each country's optimal action depends on its belief about the other's action and that belief depends on the other's cost. In lines 20–22, I set the number of states, actions, and

countries. In line 25, I collapse the state pair into a single index $j = 3s_R + s_W + 1$ running from 1 to 9. In line 28, I preallocate the $9 \times 3 \times 2$ array of probabilities. A state with no observations would give `0/0` at line 40 and so return `NaN` rather than a number. The loop in lines 30 to 44 runs over the nine states, the two countries, and the three actions. For each state, line 33 forms a logical index `instate` identifying the periods in which the state equals j. Line 40 then computes the empirical probability as the number of periods in that state in which country i chose action k, divided by the total number of periods in that state. Back in `PS2_Q3.m`, lines 26–28 invert the state index to recover the (s_R, s_W) labels for the table, and lines 30–31 count the observations in each state using `accumarray`. Lines 37–38 print this as the table `res3`.

`res3 = 9×10 table`

	State	s_R	s_W	N	R_a0	R_a1	R_a2	W_a0
1	1	0	0	30	1	0	0	0.0333
2	2	0	1	34	0	0.8529	0.1471	0
3	3	0	2	66	0.1212	0.5152	0.3636	0.0606
4	4	1	0	109	1	0	0	0
5	5	1	1	79	0	0.6582	0.3418	0
6	6	1	2	9	0	1	0	0
7	7	2	0	23	0.3478	0.6522	0	0.5217
8	8	2	1	19	0	0.4737	0.5263	0.3158
9	9	2	2	5	0	0	1	0

4. GMM

a) Moment condition

Define observed choice indicator as $y_{ikt} = 1\{a_{it} = k\}$ and model predicted probability $\sigma_i(k \mid s_t \widehat{\Phi^e}, \theta)$. Then at the true θ, $\mathbb{E}[y_{ikt} - \sigma_i(k \mid s_t, \widehat{\Phi^e}, \theta) = 0$. With only this unconditional moment, we have (# agents) x (# actions, drop 1 for collinearity) = 4 moments which means we are underidentified to estimate 6 parameters. Thus we require an even stronger conditional moment: $\mathbb{E}[y_{ikt} - \sigma_i(k \mid s_t, \theta) \mid s_t = s] = 0$ for every s. The sample analogue is $\frac{1}{T}\sum_{t=1}^{T} 1\{s_t = j\}(y_{ikt} - \sigma_i(s_t, \widehat{\Phi}; \theta)) = 0$ for each i, k, j. This gives moments (# agents) x (# actions, drop 1 for collinearity) x (# of states) = 36.

b) Estimate parameters

In `PS2_Q4.m` lines 20–42 I estimate θ by GMM. In lines 21–25, I estimate the first-stage choice probabilities with `ccp_nonparam.m` and count the observations in each state. The process is built in two subprograms. In `sigma_model.m`, I form the model-predicted choice probabilities for a candidate θ. In lines 14–15, I compute each country's expected action in each state, which replaces the sum over the rival's actions because the payoff vector is linear in a_{-i}. The loop in lines 22 to 43 runs over the nine states, two countries, and three actions. In lines 24–28, I recover (s_R, s_W) from the state index and pick out the country's own cost

state and its rival's expected action. In lines 30–37, I build the vector $\bar{\Phi}_i(a_i, s)$ of payoff determinants, with a separate branch for $a_i = 0$ in which only the β_{3i} term is nonzero. In line 39, I form $\bar{\Phi}_i(a_i, s)'\theta$ for each action, and in line 41, I convert the three utilities into probabilities using the extreme value logit formula.

In `gmm_obj.m`, I form the minimization criterion. In line 14, I subtract the model-predicted probabilities from the empirical ones, and in line 17, I weight each state's residuals by N_j/T, so each moment is $\frac{N_j}{T}(\hat{\sigma}_i(k \mid j) - \sigma_i(k \mid j, \theta))$. In line 20, I drop the residual for action 0 in each country and state, since the three probabilities sum to one and the third residual is redundant, leaving $2 \times 2 \times 9 = 36$ moments. Since the model is over-identified with $q = 36 > p = 6$, the identity matrix used for weighting is not efficient, so the estimator is consistent but not the most efficient available.

Back in `PS2_Q4.m`, line 29 wraps `gmm_obj.m` into a function of θ alone. Because `fminsearch` finds only a local minimum, the loop in lines 31 to 41 restarts it from 20 random starting values and keeps the estimate attaining the lowest criterion value, which line 42 reports.

`ans` = 6×2 table

	Parameter	GMM Estimate
1	'beta0'	-0.8130
2	'beta1R'	-0.0719
3	'beta1W'	-2.0589
4	'beta2'	-2.4101
5	'beta3R'	-2.4538
6	'beta3W'	-2.5746

Math/logic for GMM, as I worked through it:

$U_{0i}(a_i, a_{-i}, s_i; \theta) = \phi_i(a_i, a_{-i}, s_i)'\theta$

$$\phi_R(a_R > 0, a_W, s_R) = \begin{pmatrix} -a_R \\ -a_R a_W \\ 0 \\ -(2 - a_R)s_R \\ 0 \\ 0 \end{pmatrix}, \quad \phi_R(a_R > 0, a_W, s_R) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -a_W \\ 0 \end{pmatrix}, \quad \phi_R(a_R > 0, a_W, s_R) = \begin{pmatrix} -a_W \\ 0 \\ -a_R a_W \\ -(2 - a_W)s_W \\ 0 \\ 0 \end{pmatrix},$$

$$\phi_R(a_R > 0, a_W, s_R) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -a_R \end{pmatrix}$$

$$\bar{a}_{-i}(s) = \sum_{k=0}^{2} k \cdot \hat{\sigma}_{-i}(k \mid s)$$

Moments are $g_{ikj}(\theta) = \frac{N_j}{T}(\hat{\sigma}_i(k \mid j) - \sigma_i(k \mid j, \theta))$.

c) Bootstrap SEs

In `PS2_Q4.m` lines 44–82 I bootstrap the standard errors. In lines 45–47, I set the number of replications to 200, preallocate a 6×200 matrix to hold the estimates, and initialize a counter for discarded resamples (ones where a state probability was zero). In line 49, I set the seed to 1 so the results are reproducible and do not collide with the seed used for the multistart in Problem 4b. The loop in lines 51 to 79 runs one replication at a time. In line 56, I draw T row indices from 1 to T with replacement by scaling `rand` by T and taking the ceiling, and in lines 57–58 I use those indices to form the resampled data. In line 61, I re-estimate the choice probabilities on the resampled data. In lines 54–67, I wrap the draw in a `while` loop that discards any resample in which some state cell is empty, since `ccp_nonparam` returns `NaN` in that case and the `NaN` propagates through the objective. This is necessary because state 9 contains only 5 of the 374 periods in the actual data. In lines 69–70, I recompute the state index and the cell counts on the resampled data. In line 73, I build a new objective pointing at the resampled choice probabilities and counts, and in line 75 I minimize it starting from the full-sample estimate $\hat{\theta}_{GMM}$. After the loop, line 80 takes the standard deviation across the 200 replications for each parameter, which gives the bootstrap standard errors below.

`ans = 6×3 table`

	Parameter	GMM Estimate	Bootstrap SE
1	'beta0'	-0.8130	0.2093
2	'beta1R'	-0.0719	6.0027
3	'beta1W'	-2.0589	7.6121
4	'beta2'	-2.4101	8.5473
5	'beta3R'	-2.4538	11.9897
6	'beta3W'	-2.5746	16.6496

d) Interpret results

Only β_0 is estimated with any precision. Its estimate of -0.813 with a standard error of 0.209 is more than three standard errors from zero. Since own emissions enter utility as $-\beta_0 a_i$, a negative β_0 means that emitting raises utility by roughly 0.81 utils per unit. This might initially seem counterintuitive, but is consistent with pollution being a by-product of valuable production.

None of the remaining five parameters is distinguishable from zero. The point estimates of β_{1R} and β_{1W} are both negative, which as discussed earlier indicates that the two countries' emissions are strategic complements. However with standard errors of 6.00 and 7.61 neither estimate can conclusively support that. The estimate of β_2 is negative, implying that a higher unit cost of abatement leads a country to emit less This is the opposite of predicted, however again the standard error is large enough that the sign carries no information. The same is true of β_{3R} and β_{3W}, whose negative estimates would imply that a country benefits from its

neighbor's pollution in periods when it emits nothing. The imprecision traces back to the choice probabilities in Problem 3; as several are exactly zero or one, and a logit with finite parameters cannot reproduce a degenerate probability, the objective is nearly flat in those directions. The two largest standard errors from the bootstrap correspond to β_{3R} and β_{3W}, which enter utility only in the branch where a country emits nothing; note that White emits nothing in only 23 of the 374 observed periods.

5. MLE

a) Negative log likelihood

Log likelihood function: $L(\theta) = \sum_i \sum_t \ln(\sigma_i(a_{it} \mid s_t, \breve{\Phi}^e, \theta))$

In `nll_obj.m`, I form the minimization criterion. In line 14, I compute the model-predicted probabilities at the candidate θ. In line 17, I build the observed counts by multiplying each state's choice probabilities by the number of periods in that state, so that each cell holds the number of times country i chose action k in state j. This replaces the sum over the 374 periods, since the model probability is constant within a state. In line 20, I multiply each count by the log of its model probability and sum over all states, actions, and countries, and in line 21 I negate the result so that it can be passed to a minimizer. Unlike the GMM criterion, no action is dropped here, since the likelihood uses the probability of every observed choice.

`res5 = 6×3 table`

	Parameter	Estimate	Bootstrap_SE
1	'beta0'	-0.6102	0.1250
2	'beta1R'	0.4957	0.1843
3	'beta1W'	-0.1326	0.2078
4	'beta2'	-0.7089	0.1223
5	'beta3R'	-0.3676	0.2348
6	'beta3W'	0.1433	0.3622

b) Estimate parameters

In `PS2_Q5.m` lines 24–46 I estimate θ by MLE. In lines 26–29, I estimate the first-stage choice probabilities with `ccp_nonparam.m` and count the observations in each state. The process is built in two subprograms. `sigma_model.m` is unchanged from Problem 4, and `nll_obj.m` is described in Problem 5.a) above. Line 33 of PS2_Q5.m wraps `nll_obj.m` into a function of θ alone, holding the data fixed. Because `fminsearch` finds only a local minimum, the loop in lines 35 to 45 restarts it from 20 random starting values and keeps the estimate attaining the lowest criterion value, which line 46 reports.

`ans = 6×2 table`

	Parameter	MLE Estimate
1	'beta0'	-0.6102
2	'beta1R'	0.4957
3	'beta1W'	-0.1326
4	'beta2'	-0.7089
5	'beta3R'	-0.3676

	Parameter	MLE Estimate
6	'beta3W'	0.1433

c) Bootstrap SEs

In `PS2_Q5.m` lines 48–84 I bootstrap the standard errors. In lines 49–51, I set the number of replications to 200, preallocate a 6×200 matrix to hold the estimates, and initialize a counter for discarded resamples (ones where a state probability was zero). In line 53, I set the seed to 1, the same seed used in Problem 4c, so that both estimators are bootstrapped on identical resamples and any difference in the standard errors reflects the estimator rather than the draws. The loop in lines 55 to 81 runs one replication at a time. In line 60-62, I draw T row indices from 1 to T with replacement and use those indices to form the resampled data. In line 65, I re-estimate the choice probabilities on the resampled data. In lines 58–71, I wrap the draw in a `while` loop that discards any resample in which some state cell is empty, since `ccp_nonparam` returns `NaN` in that case and the `NaN` propagates through the objective. This is necessary because state 9 contains only 5 of the 374 periods in the actual data. In lines 73–74, I recompute the state index and the cell counts on the resampled data. In line 77, I build a new objective pointing at the resampled choice probabilities and counts, and in line 78 I minimize it starting from the full-sample estimate $\widehat{\theta}_{MLE}$. After the loop, line 84 takes the standard deviation across the 200 replications for each parameter, which gives the bootstrap standard errors below.

`ans = 6×3 table`

	Parameter	MLE Estimate	Bootstrap SE
1	'beta0'	-0.6102	0.1250
2	'beta1R'	0.4957	0.1843
3	'beta1W'	-0.1326	0.2078
4	'beta2'	-0.7089	0.1223
5	'beta3R'	-0.3676	0.2348
6	'beta3W'	0.1433	0.3622

d) Interpret results

All six parameters are estimated far more precisely than under GMM. Three are distinguishable from zero. The estimate of β_0 is -0.610 with a standard error of 0.125, nearly five standard errors from zero, and as in Problem 4 the negative sign means that emitting raises utility by roughly 0.61 utils per unit. Again this is consistent with pollution being a by-product of valuable production. The estimate of β_{1R} is $+0.496$ with a standard error of 0.184, so it is positive and significant, meaning that the two countries' emissions are strategic substitutes from Red's perspective: Red emits less when it expects White to emit more. The estimate of β_{1W} is -0.133 with a standard error of 0.208 and is indistinguishable from zero, so White does not show a comparable substitution response. The game is therefore asymmetric. The estimates of β_{3R} and β_{3W} are also indistinguishable from zero, which is again unsurprising since these parameters enter utility only in the branch where a country emits nothing, and White emits nothing in only 23 of the 374 observed periods.

The estimate of β_2 is -0.709 with a standard error of 0.122, so it is significantly negative, implying that a higher unit cost of abatement leads a country to emit less. This is the opposite of what the model predicts and, unlike

in Problem 4, the standard error is now small enough that the sign cannot be dismissed as noise. My theory is that this may be an artifact of the functional form: because the $a_i = 0$ branch carries no abatement cost term, a positive β_2 would push a country away from $a_i = 1$ toward both $a_i = 0$ and $a_i = 2$ as s_i rises.

The MLE standard errors are an order of magnitude smaller than the GMM standard errors, ranging from 0.12 to 0.36 against 0.21 to 16.65. Because both were bootstrapped on the same 200 resamples, the difference reflects the estimator rather than the draws. This is the efficiency tradeoff described in class---maximum likelihood is more efficient when the distributional assumptions hold, while GMM imposes fewer assumptions on the distribution of the error terms and may yield less efficient estimates. Here the problem specifies that the error terms are distributed extreme value, so the logit likelihood is likely correctly specified and thus MLE is more efficient.

6. Questions to answer with structural model

The model recovers unobserved economic parameters, allows counterfactual simulation, and enables welfare calculation. The parameters answer questions that cannot be read off the data directly. The MLE estimate of β_{1R} is positive and significant, which suggests that Red treats its own emissions and White's as strategic substitutes and emits less when it expects White to emit more. The estimate of β_{1W} is not distinguishable from zero, so White shows no comparable response. Hence the model indicates that the two countries do not respond to each other symmetrically. The estimates of β_{3R} and β_{3W} were intended to measure the externalities of damage each country suffers from its neighbor, but both are insignificant, so this particular question is one the model poses clearly without the data being able to answer it.

Once θ is estimated, the model can also be re-solved under states that were never observed. Using the estimates from Problem 5, we could ask what emissions would be if a subsidy lowered s_R by one level, and the answer would include not only Red's direct response but White's strategic response to it. A reduced-form regression of emissions on costs could not answer this, since its coefficient already confounds the direct effect with the equilibrium response and would not be stable under the policy change. Because $U_i(a, s, \varepsilon_i; \theta)$ is estimated for both countries, we can also compare equilibrium welfare to what a social planner internalizing the externality would achieve, which could inform the development of a pollution treaty that would benefit both countries.

7. Additional state variables

The model conditions only on the two abatement costs, so everything else is absorbed into the error term. The estimates suggest this may be a costly simplification. For instance, the strongly negative β_2 implies that a country emits less when abatement becomes more expensive, which is the opposite of what theory predicts. this may indicate that the specification is missing something the cost state is standing in for.

The most useful addition would be a measure of how much pollution actually crosses the border, such as prevailing wind direction. The transmission rate is currently folded into β_{1i} and β_{3i} and treated as constant, when realistically it would period to period with weather. Then the strategic response found for Red should be strongest in periods when White's pollution is drifting over. In addition, income or output states would capture the scale of the polluting activity and the willingness to pay for environmental quality, and population density would capture exposure to damage. These would also let β_0 and β_2 vary across countries to relax the assumption that Red and White have identical own-pollution benefits and sensitivity to abatement cost.

However, note that including more state variables would require either substantially more data or a parametric first stage in place of the nonparametric empirical frequencies.

II. Ford & GM

Wald, Matthew. (2007). Bush tours 2 auto plants and talks of energy policy. New York Times, 21 March 2007. Brief summary: President Bush toured Ford and GM plants and pushed ethanol, hybrids and fuel cells. The article says the Detroit firms have “fallen behind in the hybrid business, which is now dominated by Toyota,” and that the administration de-emphasized hybrids in favor of hydrogen fuel cells. The Claycomo plant already makes hybrid Escapes and Mariners, which qualify for tax credits.

8. Decision-making problem faced by Form & GM as a static game

a) Research question

Does Ford’s and GM’s choice of which new powertrain to commit to depend on what each expects the other to choose? If so, is the interaction one of strategic complements or substitutes, and does it differ by technology?

Ethanol and hydrogen need shared fueling infrastructure. If one firm commits, the other’s payoff from the same technology should rise (complements). For hybrids, where Toyota already dominates, a second Detroit entrant mainly splits a thin margin (substitutes). Whether infrastructure complementarity or business stealing wins for each technology is an empirical question. A single-firm model cannot answer it, and it determines how policy such as tax credits, clean fuel subsidies, or fuel-cell R&D support will work.

b) Decisions a_i

There are two players, $i \in \{F, G\}$. In each period each firm chooses one new powertrain commitment, $a_i \in \{0, 1, 2, 3\}$, where 0 = conventional gasoline, 1 = ethanol, 2 = hybrid, and 3 = hydrogen fuel cell. The action profile is $a = (a_F, a_G)$, and a_{-i} is the rival’s action. Gasoline is the baseline, meaning no new commitment.

c) State variables s_i

The publicly observed state is $s = (s_F, s_G, s_0)$. Firm-specific components s_i are common knowledge to both firms and observable to the econometrician, which include:

- truck_i, the share of the firm’s lineup in trucks and SUVs where gasoline is most common,
- legacy_i, the firm’s legacy labor costs (retiree health care and pension obligations) divided by annual vehicle production, a measure of how financially constrained the firm is,
- capability_i, existing capability in the technology (hybrid and flex-fuel models already offered, battery and R&D stock),
- share_i, the firm’s market share.

Market-level components s_0 are common to both firms and include gas prices, federal hybrid tax credit and ethanol subsidy, CAFE standard stringency, count of E85 and hydrogen stations, Toyota’s hybrid share (the article’s key competitor), and a political signal about which technology the administration favors.

d) Strategic interaction + externalities

The firms interact through four main channels. Each firm's payoff from a technology depends on the other's choice, so this is a game and not two separate single-agent problems. Each of these is an externality that affects the other agent in the game:

1. **Network (infrastructure) externality:** consumers buy an ethanol or fuel-cell car only if fueling stations exist, and stations are built only if enough such cars are on the road. One firm's commitment raises the value of the same technology to the other. This makes the choices strategic complements for technologies 1 and 3.
2. **Competition:** if both firms enter the same product with the same technology, they split demand and lower the price. For hybrids, where Toyota already holds the market, a second and third entrant compress margins. This makes the choices strategic substitutes.
3. **Supply spillovers:** battery, fuel-cell and ethanol supply chains are shared. A rival's investment lowers a firm's own cost of the same technology.

There is also an environmental externality: reduced oil dependence and lower emissions benefit parties outside the game. This does not create strategic interaction, but it is the reason the government subsidizes certain technologies and the reason the planner cares about the equilibrium.

e) Utility function

Firm i's period payoff from action a_i given the rival's action a_{-i} is:
$U_i(a_i, a_{-i}, s, \varepsilon_i; \theta) = \pi_i(a_i, a_{-i}, s; \theta) + \varepsilon_i(a_i)$. Normalize gasoline to $\pi_i(0, a_{-i}, s) = 0$. Then for $k \in \{1, 2, 3\}$ the payoff is $\pi_i(k, a_{-i}, s) = \beta_{0k} + s_t'\beta_k + s_0'\gamma_k + \sum_{m=1}^{3} \delta_{km} \mathbf{1}\{a_{-i} = m\}$.

Interpretation: $\pi_i(k, \cdot)$ is the profit from committing to technology k relative to staying with gasoline. β_k measures how firm-specific traits change that payoff. I expect truck share to lower the payoff from hybrids and fuel cells (gasoline trucks are the profit center), and existing capability to raise it. γ_k measures how market conditions matter---higher gasoline price and higher subsidies should raise the payoff. δ_{km} is the strategic parameter, the change in the payoff to technology k when the rival chooses m. The diagonal terms δ_{kk} test the research question: $\delta_{11}, \delta_{33} > 0$ means infrastructure complements, and $\delta_{22} < 0$ means hybrid business competition. Off-diagonal terms capture cross-technology effects, such as a rival's ethanol push making the hybrid less attractive (shared infrastructure/supply).

The shocks $\varepsilon_i(a_i)$ are private information, observed only by firm i before acting. They are assumed i.i.d. across players and actions, independent of s, and Type I extreme value. Private information makes the equilibrium a Bayesian Nash equilibrium in beliefs. Independence across players rules out unobserved common shocks. The extreme value assumption gives closed-form choice probabilities. With $P_j(m|s)$ the probability that the rival chooses m, firm i's choice probability is

$P_i(k|s) = \dfrac{\exp(v_{ik}(s))}{1 + \sum_{l=1}^{3} \exp(v_{il}(s))}$, $\quad v_{ik}(s) = \beta_{0k} + s_t'\beta_k + s_0'\gamma_k + \sum_{m=1}^{3} \delta_{km} P_j(m|s)$. The parameters to be estimated are $\theta = (\beta_{0k}, \beta_k, \gamma_k, \delta_{km})$ for $k, m \in \{1, 2, 3\}$.

f) Data

The unit of observation is a firm-product-year, and one "game" is a product-year in which both firms choose. Possible sources:

- **Powertrain offerings:** EPA/DOE Fuel Economy Guide (fueleconomy.gov) lists every model-year vehicle by fuel type and hybrid status, and Wards Automotive gives model-level launches.
- **Sales and shares:** Polk/IHS registrations or Wards sales by firm and product.
- **Infrastructure:** DOE Alternative Fuels Data Center for counts of E85 and hydrogen stations by year.
- **Prices and policy:** EIA gasoline prices, IRS records of the hybrid tax credit, NHTSA CAFE standards.
- **Firm financials:** 10-K filings from Ford and GM for legacy costs, R&D, and truck share.
- **Competitor:** Toyota's hybrid sales as a market-level state.

g) Estimation and interpretation

Following the semiparametric approach in Bajari, Hong, Krainer, and Nekipelov (2010), as in class:

Step 1: Estimate each firm's conditional choice probabilities $\widehat{P}_i(k|s)$ with a flexible multinomial logit of the observed choice on the state.

Step 2: Under the extreme value assumption, the log-odds are linear in the parameters once rival probabilities are replaced by their estimates:

$$\ln \frac{P_i(k|s)}{P_i(0|s)} = \beta_{0k} + s_t'\beta_k + s_0'\gamma_k + \sum_m \delta_{km}\widehat{P}_j(m|s)$$

Estimate θ by GMM on these equations, stacked over players and technologies. Standard errors come from a bootstrap that resamples whole games and repeats both steps, which accounts for the estimated regressors from step 1. Identification requires variables that shift one firm's payoff but not the other's (instruments), such as Ford's truck share or GM's legacy costs. These affect the rival only through the rival's belief about a_i, which separates strategic effects from common shocks.

Interpretation: the signs and sizes of the δ_{kk} answer the research question. A positive δ_{11} means a rival's ethanol commitment makes the firm more likely to commit too, which is a coordination story. A negative δ_{22} means hybrid entry is deterred by the rival's entry. Magnitudes are interpretable against γ_k. For example, $\delta_{11}/\gamma_{\text{gas price}}$ is the gasoline price change that would offset the infrastructure benefit of one rival's ethanol commitment.

h) Assumptions

Simultaneity: Neither firm observes the other's choice before committing. This is roughly realistic, since product plans are private, but announcements and the White House meetings in the article provide signals, so the game may be closer to sequential.

Independence of private shocks: Ford and GM face common unobserved shocks, such as gasoline price expectations or lobbying outcomes, and these may be correlated across firms. Correlated shocks would be misread as complementarity and bias δ_{kk} upward. I would add year effects and test sensitivity to the market controls.

Single action per period. Real firms run portfolios across many models, so collapsing them to one choice per segment-year loses information. A multi-product version would be more realistic.

Linear, additive payoffs and a single-period horizon. These are standard but not obviously true. For example, the infrastructure effect probably depends on how many stations already exist, which would make it nonlinear.

i) Setup evaluation

The static game fits the within-period strategic response, meaning whether a firm leans toward a technology when it expects the rival to do the same. The setting, though, is almost certainly dynamic. Plant technology changes and R&D are costly and hard to reverse. Infrastructure builds up over time, and learning-by-doing makes today's commitment lower tomorrow's cost, sorival's commitment today changes the state of the world next year. A static model mixes the within-period strategic effect with expectations about future play. The article also shows firms reacting to a policy path (the administration's energy program and the CAFE debate), which makes timing and expectations central. A dynamic game with investment and an evolving infrastructure state would be a better model but would require far more data.

j) Questions answered from structural model

Estimating this model could allow us to draw several interesting conclusions and consider a number of counterfactuals, including the following ideas:

Recover unobserved parameters: The central one is δ_{km}, the strategic response. These cannot be recovered by a descriptive regression because Ford's and GM's choices are correlated both through strategic interaction and through common shocks. The exclusion restrictions in the estimation separate the two.

Policy simulation: We could assess the potential impacts of a larger hybrid tax credit, an E85 station subsidy, tighter CAFE standards or fuel-cell R&D support. Each of those policies shifts each firm's payoff and firms then re-optimize against each other. With complements, the equilibrium response is amplified, and a small push from the government could tip both firms to ethanol. With substitutes, the response is dampened.

Coordination failure: The model could test whether the firms are stuck in a low-adoption equilibrium that a one-time subsidy could move, which is the case for infrastructure support.

Welfare: Combining the equilibrium choices with external estimates of emissions and oil-security damages could show whether the market pace of adoption is too slow from a social standpoint (and hence reflect on potential welfare-optimizing policy).

III. Static Games

9. Example of strategic game in the power generation sector

a) Strategic decision-making problem

The problem I have chosen is the decision by owners of coal-fired generating capacity of whether to cease burning coal, taken simultaneously by competing firms within the same electricity market. In any given year, each firm owning coal units in a transmission zone (an area smaller than ISO or operating region) decides whether to keep those units burning coal or to remove them from coal service either by retiring them or by repowering them to natural gas. The firms sell into a common wholesale market, so one firm's exit shrinks supply and raises the price its rivals receive, and thus each firm's best choice is dependent on what it expects its rivals to do.

I have chosen this problem for several reasons. First, it is the setting of my ongoing research, so I have the institutional detail and the data already (and I find it interesting). Second, the strategic interaction is clear--the channel runs through a market price that is observable in a market with clearly defined administrative boundaries. Third, the question is policy relevant since the pace at which coal capacity leaves the system is a central determinant of future power-sector emissions.

b) Background and motivation

The U.S. coal fleet is long-lived capital whose operating economics have deteriorated sharply over the past fifteen years, as abundant shale gas and falling renewable costs compressed coal-fired operating margins and pushed much of the fleet toward the margin of profitability (Davis, Holladay, and Sims 2021). Most of the remaining units were built decades ago and many were grandfathered under vintage-differentiated environmental regulation, which has extended their economic lives well beyond what a level regulatory playing field would have supported (Stavins 2005; Bialek, Gregory, and Revesz 2025).

The existing economics literature on coal exit has largely treated the firm's problem as a single-agent one: a unit retires, remains (and complies with environmental policy), or converts according to its own costs and the policy environment it faces (Gowrisankaran, Langer, and Zarek 2024; Gowrisankaran, Langer, and Reguant 2025). That is a reasonable simplification when each unit is small relative to its larger market. It is less reasonable in a concentrated transmission zone, where a handful of firms own most of the coal capacity and each knows that its own exit will move the price the others receive.

If exits are strategic substitutes, that is, if each firm is less willing to leave when it expects others to leave, then the aggregate response to a carbon price or a retirement subsidy is smaller relative to what a single-agent model predicts. A policy calibrated on single-agent estimates would then overstate potential emissions reductions. If on the other hand exits are strategic complements driven by shared fixed infrastructure or by regulatory pressure that intensifies as peers depart (see Trump administration's orders for coal plants to remain running), the policy response would be amplified instead. Which of these holds is an empirical question not answerable without a model in which firms' choices depend on one another.

c) Characterization and agent

The decision-making agents are the parent firms that own coal-fired generating capacity within a given transmission zone. These could take the form of investor-owned utilities, publicly owned power authorities, or merchant generators. I take the parent firm rather than the individual unit as the player because exit decisions are made and announced at the corporate level and are typically coordinated across units at a site.

In a given year, the decision is whether to stop burning coal at the firm's coal capacity in that zone. The firms choose simultaneously, each knowing the publicly observable characteristics of all the units in the zone (size, age, utilization, ownership). Firms do not know the private considerations driving its rivals' choices in that particular year, such as internal budgets, boiler condition, and negotiation considerations with regulators or gas suppliers. Each firm therefore chooses a best response given its beliefs about the probability that each rival exits. The relevant equilibrium concept is Perfect Bayesian Equilibrium consisting of a profile of strategies in which every firm's choice is optimal given correct beliefs about the distribution of its rivals' choices.

d) Static game

This setting describes a game involving strategic interactions because the actions of one agent affect the payoffs of another in multiple ways:

1) electricity within a transmission zone clears at a common price, so when a rival withdraws capacity the residual demand facing the remaining units rises, and with it the price and the operating margin those units earn. A firm's payoff to continuing with coal is therefore strictly higher when its rivals have exited.

2) as capacity leaves a zone, the units that remain become more likely to be designated as necessary for reliable operation, which constrains their owners' freedom to exit later.

In addition, this is a static game because within a given year, the firms effectively choose simultaneously. Firms do not observe rivals' current-year decisions before committing to their own, so no firm can condition this year's action on this year's realized profile. Payoffs are realized within the period, each firm moves once, and the model takes the state of the world as given rather than tracking how today's choices shape tomorrow's.

e) Research question(s)

Do coal-plant owners' exit decisions respond to their rivals' expected exit decisions within the same transmission zone? If so, is the interaction one of strategic substitutes? (Does an expected wave of rival exits deter a firm's own exit by raising the margin it earns from staying or incentivize it to exit sooner for fear of being prevented from doing so by regulators if it remains as crucial capacity?)

f) Decisions

Each player i chooses a binary action $a_i \in \{0, 1\}, i = 1, ..., N$, where $a_i = 1$ if firm i ceases burning coal at its capacity in the zone during the period and $a_i = 0$ if it continues to burn coal for another period. The action profile is $a = (a_1, \dots, a_n) \in A = \{0, 1\}^n$, and a_{-i} denotes the actions of all players other than i.

Note that pooling retirement and repowering into a single "exit from coal" action is a simplification. The two 'fates' differ in ways that matter for other questions, but they are equivalent with respect to the strategic channel at issue here.

g) States

The publicly observed state is $s = (s_1, \dots, s_n) \in S = \prod_i S_i$, where s_i includes the characteristics of firm i's coal position in the zone. These elements are common knowledge to the players and observable to the econometrician:

- Cap_i, the firm's coal capacity in the zone,
- Age_i, the capacity-weighted vintage,
- $Util_i$, its recent utilization (capacity factor or annual operating hours),
- $Crit_i$, the firm's share of total generating capacity in the zone (how critical its capacity is to grid reliability),
- R_i, an indicator for whether the firm is rate-regulated, since cost recovery differs for regulated utilities specifically.

In addition, each state s_i includes the market-level conditions facing every player in the zone in that period. These include the delivered gas-to-coal price ratio and zone demand. These are common across players.

h) Strategic interactions and externalities

The strategic interaction runs through two channels. I also identify a third externality which operates outside the game entirely but may be policy relevant.

1) **"Pecuniary externality" from supply effects:** Electricity within a transmission zone has a common residual demand curve, so a firm's exit shrinks supply and raises the price received by every remaining unit. This raises rivals' payoff to continuing and lowers their payoff to exiting, which makes the actions strategic substitutes--each firm's incentive to exit falls as the expected number of rival exits rises. The gain to remaining generators is paid by consumers (I abstract from rate-basing here) and firms internalize it only to the extent that they are large enough in the zone for their own action to move the price, which depends on regional market concentration.

2) **"Reliability externality" from regional capacity: T**he more capacity leaves a zone, the more likely the capacity that remains is to be needed for reliable operation and the more likely the next unit choosing to exit is prevented from retiring by regulators or ISO under a reliability contract. Expected exit of rivals thus alters not only a firm's current payoff but the option it holds to exit later.

3) **Environmental externality:** Emissions of carbon dioxide and local air pollutants impose costs on parties entirely outside the game which is not reflected in these firm's payoffs. This externality clearly does not generate strategic interaction among the players but it is the reason the planner cares about the equilibrium the game produces and it is what policy instruments attempt to internalize.

i) Utility function

Firm i's period payoff is $U_i(a, s, \varepsilon_i; \theta) = U_{0i}(a_i, a_{-i}, s_i; \theta) + \varepsilon_i(a_i)$ where U_{0i} is the deterministic, public info component and $\varepsilon_i(a_i)$ is a private information shock. I normalize the payoff to continuing, $U_{0i}(0, a_{-i}, s_i; \theta) = 0$ and specify the exit payoff as $U_{0i}(1, a_{-i}, s_i; \theta) = \theta_0 + \theta_1 X_i + \theta_2 MW_i + \theta_3 Age_i + \theta_4 Crit_i + \theta_5 R_i + \theta_6 \sum_{j \neq i} a_j$.

The parameters to be estimated are $\theta = (\theta_0, \theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6)'$. Each coefficient measures the effect of a state variable on the net payoff to leaving coal relative to staying. I expect $\theta_1 < 0$ (unit running more forfeits more profits), $\theta_3 > 0$ (older capacity has less continuation value), $\theta_4 < 0$ (more critical capacity faces higher barriers to exit). Then θ_6 is the strategic parameter whose sign could help answer the research question. Its interpretation is the change in exit payoff induced by an additional rival exit, so $\theta_6 < 0$ would mean exits are strategic substitutes and $\theta_6 > 0$ suggests they are complements.

The shocks $\varepsilon_i = (\varepsilon_i(0), \varepsilon_i(1))$ are assumed to be private information observed by firm i before it acts but not by its rivals or by the econometrician. These shocks are additionally assumed to be i.i.d. across players and across actions independent of the public state s and drawn from the Type I extreme value distribution. Private shocks makes the equilibrium Bayesian Nash in beliefs and independence across players rules out shocks common to all firms in a zone that the econometrician does not observe. The extreme value assumption delivers closed-form conditional choice probabilities, given the normalization $U_{0i}(0, \cdot) = 0$ then $\sigma_i(1 \mid s) = \dfrac{\exp\left(\Phi_i^e(1, s)'\theta\right)}{1 + \exp\left(\Phi_i^e(1, s)'\theta\right)}$.

j) Data and availability

The unit of observation is a firm–zone–year, and one "game" is a zone–year. I could acquire the following data, mostly pre-cleaned from PUDL:

- **Generator characteristics:** EIA Form 860 reports nameplate capacity, in-service year, prime mover, fuel, operating status, planned retirements, and planned repowering for every utility-scale generator annually,
- **Generation and fuel:** EIA Form 923 gives monthly net generation and fuel consumption by unit,
- **Ownership:** EIA Form 861 for utility identity and service territory,
- **Market:** Transmission zone definitions from ISO/RTO maps and zonal load from FERC Form 714,
- **Prices:** EIA delivered fuel prices to the electric power sector and wholesale power prices,
- **Reliability designation:** ISO deactivation notices and reliability-must-run filings from FERC and individual ISOs.

k) Estimation and interpretation

Following the semiparametric approach in Bajari, Hong, Krainer, and Nekipelov (2010), as in class:

Step 1: Estimate the conditional choice probabilities $\widehat{\sigma_j}(a_j|s)$ from the data, i.e. the empirical frequency with which firms of given observable characteristics facing given market conditions exit coal. With continuous states, I would use a flexible parametric logit for the exit indicator on the state. These estimates are then used to form the expected payoff vector $\widehat{\phi_i}^e = \sum_{a_{-i}} \phi_i(a_i, a_{-i}, x_i) \prod_{j \neq i} \widehat{\sigma_j}(a_i|s)$ which under the linear specification reduces to replacing the rival-exit count with the sum of estimated rival exit probabilities.

Step 2: Estimate θ by GMM matching predicted to observed choices. The sample moment condition is

$\frac{1}{T}\sum_{t=1}^{T}[y_t - \sigma(s_t, \phi^e; \theta)] = 0$ where $y_{it} = 1$ if firm i exits in game t and 0 otherwise

Bootstrap to get standard errors by resampling the games with replacements and reestimating on each.

Interpretation: The sign of $\widehat{\theta_6}$ answers the research question directly as mentioned earlier. Negative means strategic substitutes where firms are deterred from exiting by the expectation that rivals will exit. Its magnitude is interpretable relative to the other coefficients, for example $\widehat{\theta_6} / \widehat{\theta_1}$gives the change in utilization that would offset one additional expected rival exit.

l) Assumptions

This model's key assumptions include the following:

Simultaneity: The model assumes no firm observes a rival's current-period action before choosing. In practice retirements are announced publicly several years in advance, which points towards a sequential rather than simultaneous game structure.

Independence of private shocks: This is a standard assumption, but in this context it would not be unreasonable to think that unobserved zone-level shocks like regional regulation or local gas supply constraints could introduce correlation in exit decisions. Here this would be incorrectly attributed to the strategic parameter here (bias it towards complementarity interpretation). This could be partially controlled for or tested by included zone fixed effects and testing sensitivity of the estimates to observable market controls.

Linearity of payoffs in rival exits: . The specification assumes one additional rival exit matters as much as the fifth. If the price effect is convex in withdrawn capacity, then a specification in rival capacity rather than rival

count would be preferable. Considering that we are using zonal capacity proportion as a criticality measure, using rival capacity would likely be more realistic.

Fixed set of players: Players are assumed to be the firms holding coal capacity in the zone at the start of the period. Since exit is absorbing, the player set shrinks over time. This does not fit with the framework treating each zone-year as a draw from the same game. A better model would factor in some kind of game evolution (see below).

m) Setup evaluation

The static setup is appropriate for the research question specified originally. The research question asks about the sign and magnitude of a strategic interaction which is present within a single period. The price effect of a rival's exit is felt immediately, and the firm's best response to its beliefs about rivals is a well-defined object within the year. However, this framework abstracts from the potentially more important question about exit timing, which I discussed in Problem Set 1. Exit from coal is irreversible, and the economically interesting margin is not whether a firm would ever leave but when. A static model cannot represent the option value of deferring an irreversible decision nor the fact that a rival's exit today changes the state for the next period's decision. The static model estimation thus mixes together the within-period price effect and the firms' expectations about future play. Perhaps the ideal model would account for both of these ideas.

In addition, since exit is absorbing, a model in which the same players face the same game each period is at best an approximation to a situation in which the player set is itself an outcome. The theoretically correct object is a dynamic stopping game of the kind studied in the timing-games literature such as that of Riedel and Steg (2017). Due to these assumptions, I would assess that while the static game is a good place to start, it is important to interpret the resulting estimate as a within-period strategic response and to not conflate this with the structural parameters of the intertwined optimal stopping problem.

n) Counterfactuals of interest

Estimating this model could allow us to draw several interesting conclusions and consider a number of counterfactuals, including the following ideas:

Recover unobserved parameters: The central one is θ_6, the strategic response. Since every firm in a zone faces the same market conditions, the correlation in their exit decisions is observationally equivalent to a common shock and this cannot be recovered with a descriptive regression.

Policy simulation: A carbon price, a retirement subsidy, or a tightening of an air-quality standard (e.g., MATS) shifts every firm's payoff to exiting and the firms then re-optimize against one another. If exits are strategic substitutes, the equilibrium response is smaller than the sum of the individual responses. A single-agent model would miss this dampening entirely and overstate the emissions reduction.

Welfare estimates: Because the payoffs are in a common scale, the equilibrium outcomes can be combined with external estimates of the damages from emissions and the costs of reliability procurement to investigate whether the equilibrium pace of exit is too slow or too fast from a social standpoint. Given the environmental externality discussed earlier, this is potentially the most motivating question of interest for estimating this model.

In addition, the model provides a decomposition of observed delay. Coal units have remained in service longer than unit-level cost conditions alone would suggest. In a similar thread as the optimal stopping model

discussed in Problem Set 1, this estimated model would allow the part of delay attributable to strategic interactions to be separated from that due to firm-level idiosyncrasies.